HRLC® System Interface
Reference Manual

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Chapter 1
Introduction

1.1 About the HRLC System Interface

The HRLC system interface (SI) is an intelligent, dynamically configured device that coordinates all the HRLC system components. The SI is a small, specialized computer, which transmits information between the data station (the host computer and the HRLC software) and the system's various chromatographic instruments. When prompted by the data station, the SI controls and monitors the system instruments, transmitting back to the data station raw chromatographic data and information about system status. The activities of the SI free the data station from the task of direct instrument control, thus diminishing dependence of the chromatographic system on the host computer. This arrangement reduces the number of cables needed to connect the data station to system instruments, and allows the data station to handle the more complex tasks of data integration and communication to the user.

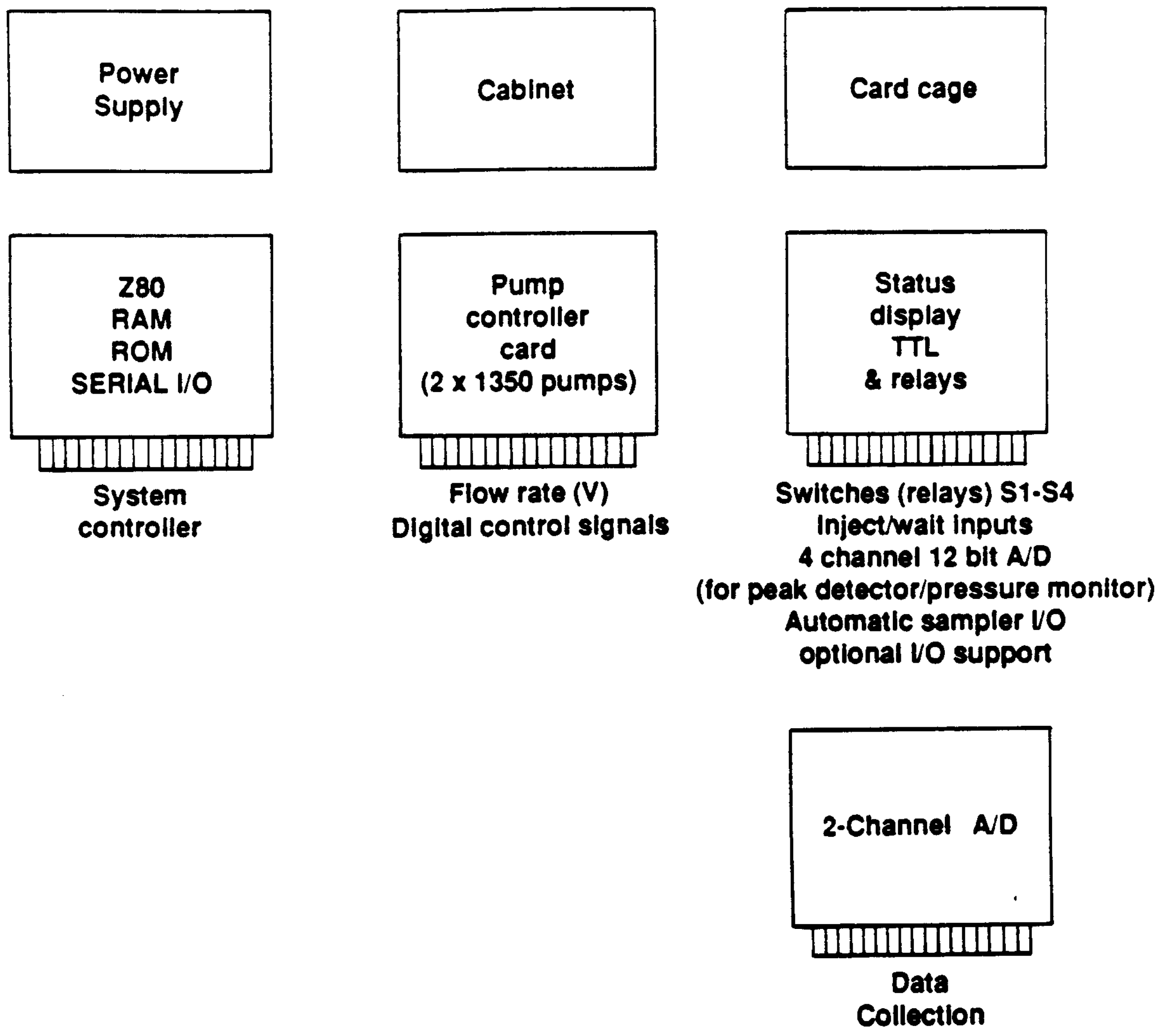

Fig. 1.1. System Interface hardware components.

The system interface consists of several function cards. Each card controls and monitors specific instruments which perform complex tasks, such as collecting data and delivering gradients, and simpler tasks, such as making injections. Using the data station and working through the cards within the SI, a user can specify which instruments to monitor and control. Figure 1.2 illustrates the role played by the SI's several function cards as information traffics between the data station, the SI, and the chromatographic equipment.

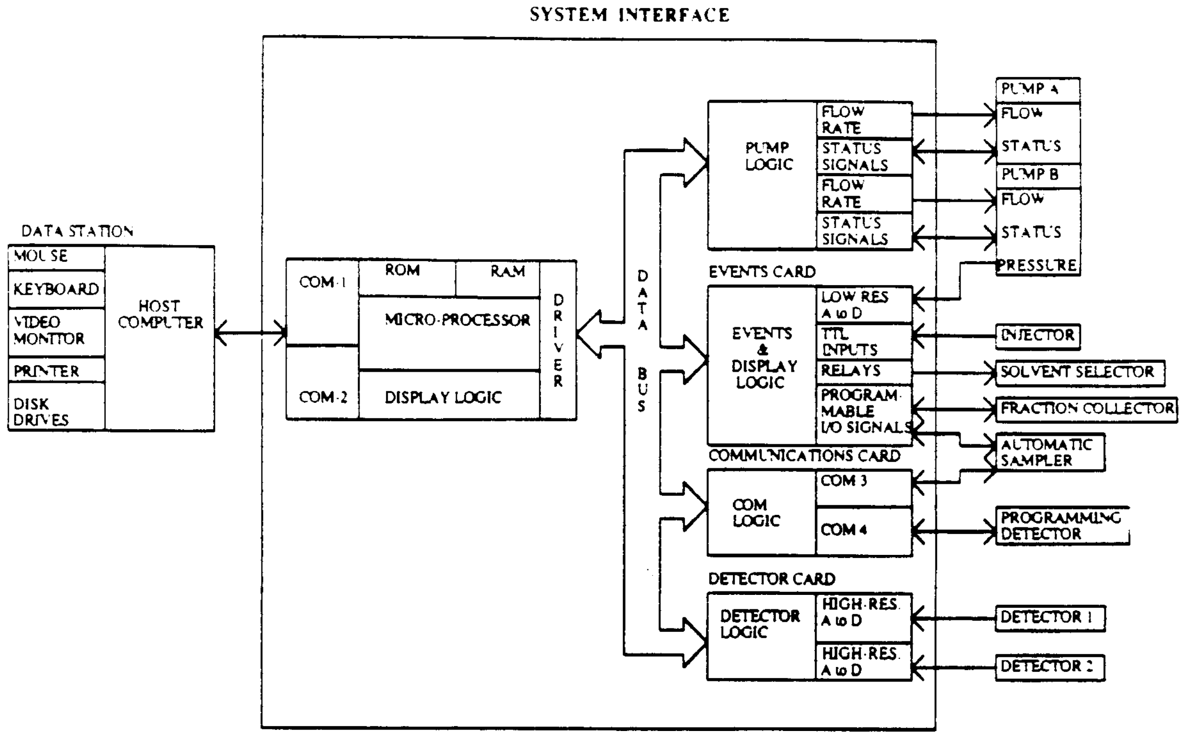

Fig. 1.2. System Interface block diagram.

The CPU card, along with the events card, is always present in each system interface. The CPU card works as a traffic coordinator. It communicates with the other function cards via the STD-bus, and with the data station via an RS-232C serial port. The CPU card directs information and messages to and from other function cards that communicate directly with the instruments being controlled. The CPU card also receives, from these function cards, information about the status of the system. The CPU card deciphers this information, taking action as required and storing the information to transmit at the appropriate time to the data station.

The events card controls the peripheral chromatographic equipment. Logical input and output signals on the events card control injectors, samplers, fraction collectors, detectors, and other instruments. A maximum of four low-resolution analog inputs on the events card monitor pressure and temperature signals, and can also be used to monitor the low-resolution chromatographic data coming from detectors.

Other function cards may be added to the SI. The pump card directs two Series 1350 Soft-Start® pumps. The card monitors the status of the pumps and relays to them the flow rates programmed within the HRLC software (at the data station). The resulting gradients produce the chromatographic separation. The detector card converts the analog data coming from the detectors into high resolution digital data, which is buffered by the CPU card and sent to the data station when the computer is ready to receive it.

The front panel of the system interface gives the user general status information about the progress of the functions being carried out by the SI. This includes information about the status of the pumps, which must otherwise be monitored using the HRLC software running on the data station. The ability to monitor the pumps from the SI's front panel is especially useful when the data station is a distance away.

1.2 External Description

This section describes the purpose of the elements that make up the front and back panels of the HRLC system interface. Figure 1.3 shows the SI front panel. It consists of several LED status lights and a local/remote switch. The front panel serves as the local user interface to the SI. A complete description of each function is found in Table 1.1.

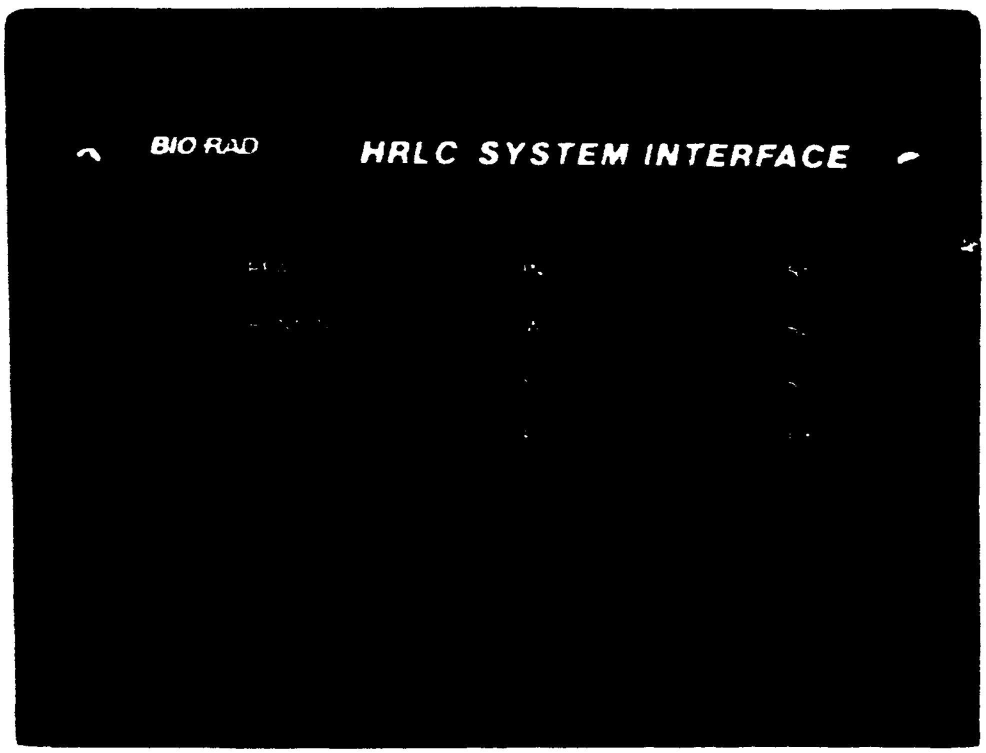

Fig. 1.3. SI front panel.

The back panel contains the connections for the various devices that may be attached to the SI. Figure 1.4 shows the basic back panel found on all SIs. It contains all the connections for one pump card, two detector cards, one CPU card, and one events card. These connections are described in Table 1.2. In the Series 800 interfaces, an additional panel replaces the blank panel shown in Figure 1.4. This panel is a pump expansion panel, providing connectors for two additional pump cards. Figures 1.5 and 1.6 illustrate these panels.

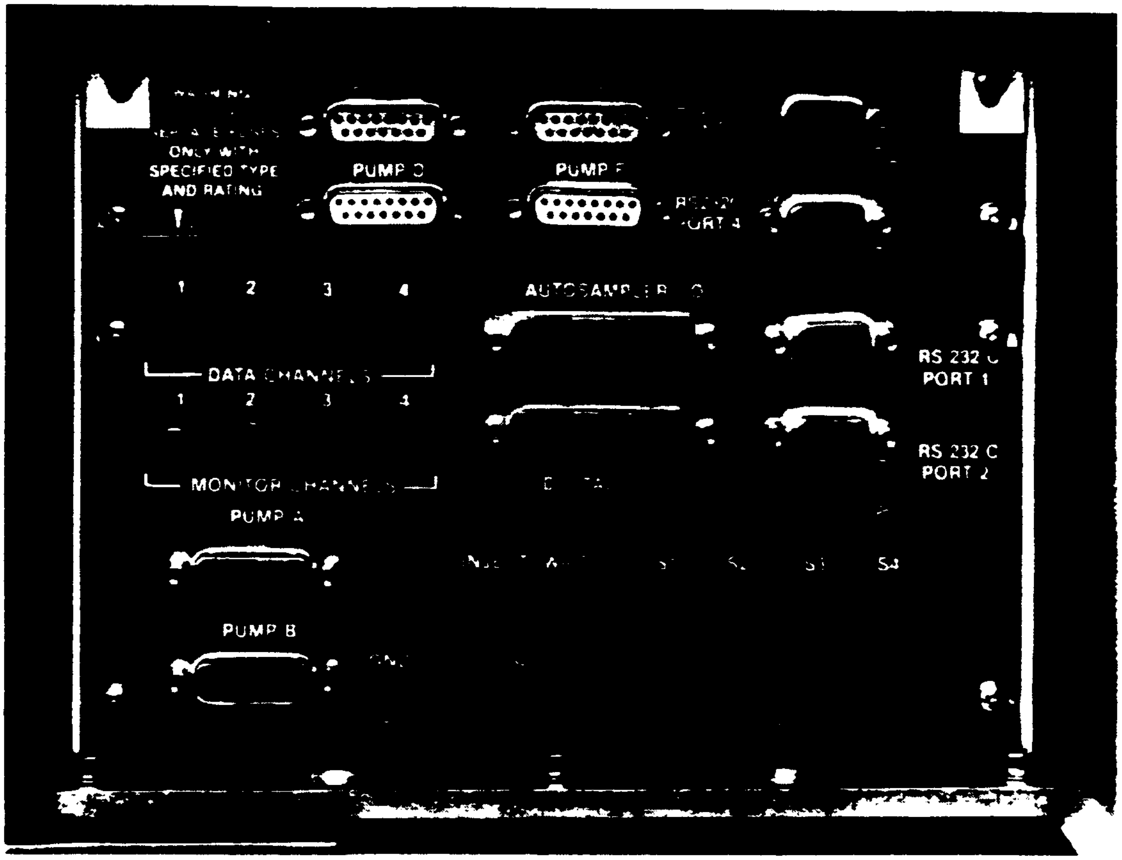

Fig. 1.4. Series 800 system interface rear panel.

The serial number label on the left hand side of the top cover indicates the unit model number and serial number. Be certain that the serial number matches that on the warranty card. The warranty card must be submitted to register your warranty and qualify you to receive any future software updates.

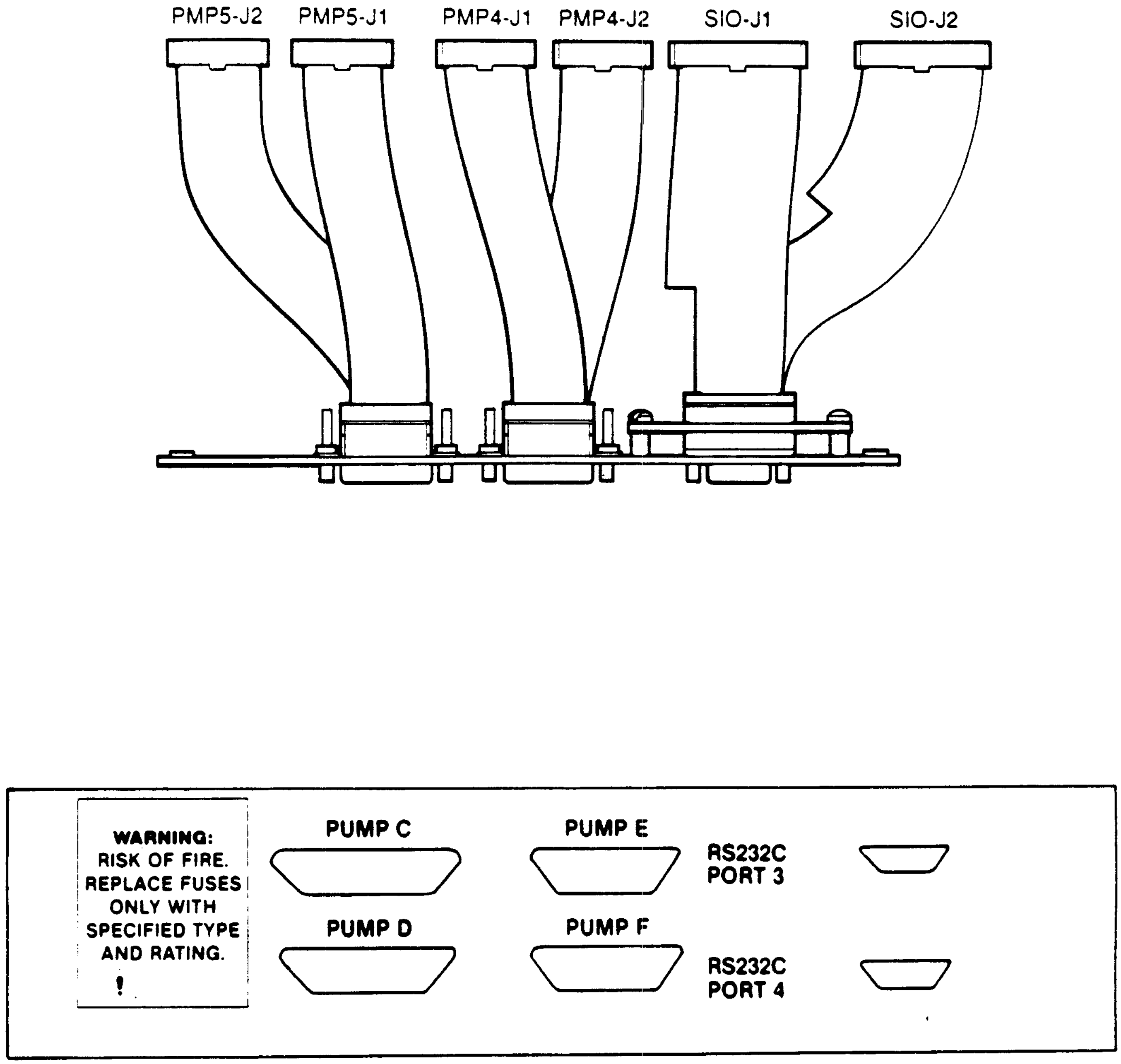

Fig. 1.5. Pump expansion panel with ribbon cables.

Table 1.1. Front and Rear Panel Components

A. Method status indicators

B. Local/Remote switch and indicators

C. RS-232C connections (Data Station Connector)

D. TTL input and output (Autosampler and Fraction Collector Connectors)

E. Relay outputs and injector inputs

F. Low-resolution analog to digital connector inputs

G. High-resolution (Integrator) analog to digital connector inputs

H. Pump control connectors

I. Power entry module containing power switch, fuses, voltage selection card, and universal power cord connector

J. Serial number plate (underneath and not shown)

Chapter 2
Installation

2.1 Unpacking

The HRLC system interface is available in two configurations: stand-alone (PC integrators) or built into an HRLC gradient module. To install a stand-alone unit, remove the unit from its shipping carton and position it near the other system components. To install a gradient module, follow the installation instructions in the gradient module reference manual. Since the system interface is already assembled within the gradient module, this system interface manual need be used for reference only.

Before attaching the power cord, refer to Section 2.2 for the correct line voltage. Save the packing materials to facilitate shipping in case repair should be required at one of our depot service centers.

Inspect the accessories. All the items you should have received with your interface are listed below. The model number describes the configuration of the SI when it left the factory. The various interface cards use different cables. Therefore, the accessory kit will vary depending on which configuration has been purchased.

When the system interface is purchased as a stand alone unit it comes with;

125-1622	Analog to digital cable
900-7253	1 amp fuse (250 V)
900-7278	2 amp fuse (250 V)

When the system interface is purchased with a gradient module it is shipped with;

125-1622	Analog to digital cable
125-1618	Manual injector cable
125-1632	Pump cable (2)
900-7253	1 amp fuse (250 V)
900-7278	2 amp fuse (250 V)

2.2 Configuring for Different Line Voltages

A built in system interface has its voltage configured for the destination country with the correct fuses in place. The voltage is listed on the serial number label on the back of the gradient module. If the gradient module is to be used with a different voltage, the SI voltage configuration will have to be changed before the system interface can be used. All stand-alone interfaces are shipped with a 120 V selection. Therefore, they will have to be re-configured for operation at another voltage if required. A 120 V selection is required for operation at 100 V or 120 V; a 240 V selection is required for operation at 220 V or 240 V.

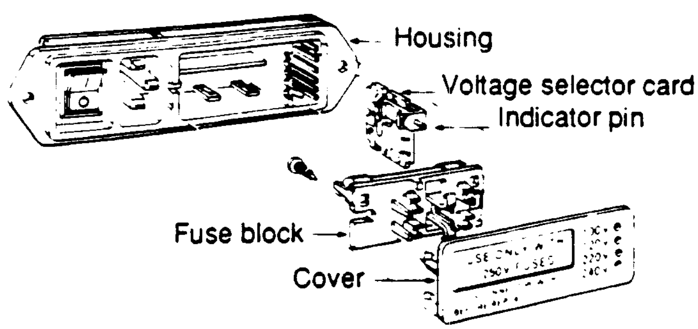

Fig. 2.1. Fuse plate, fuse block, and voltage card assembly.

2.2.1 Inserting Fuses for 100 V and 120 V Operation

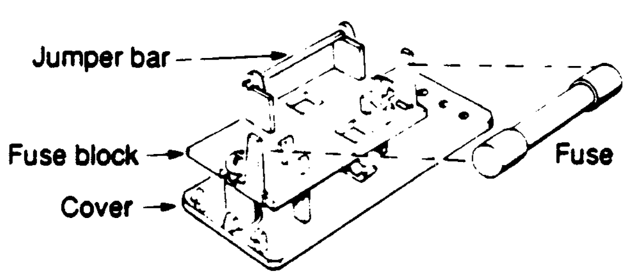

Fig. 2.2. Fuse block position and fuse insertion for 100 V and 120 V operation.

1. Open cover, using a small blade screwdriver or similar tool.

2. Insert one 2 A fuse in the fuse block as shown in Figure 2.1. The 2 A fuses are 1 1/4 inches long, fast blow.

3. Put the fuse block and cover back into the housing.

2.2.2 Inserting Fuses for 220 V and 240 V Operation

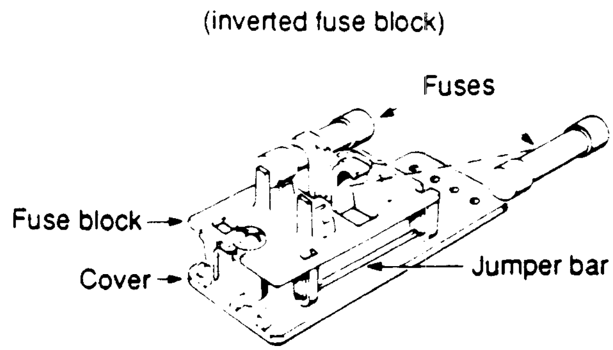

Fig. 2.3. Fuse block position and fuse insertion for 200 V and 240 V operation.

1. Open cover, using a small blade screwdriver or similar tool.

2. Loosen the Phillips screw on the fuse block one turn.

3. Remove the fuse block by sliding it up and away from the Phillips screw and lifting it up from the pedestal.

4. Invert the fuse block (see Figure 2.3 for correct orientation), and slide it back onto the Phillips screw and pedestal.

5. Retighten the screw.

6. If they are not already installed, insert the two 1 A time-delay fuses into the block as shown in Figure 2.3. These fuses are 2 cm long.

7. Adjust the voltage selection card for the appropriate voltage, following the instructions in Section 2.2.3.

8. Put the fuse block and cover back into the housing.

2.2.3 Voltage Selection Card Orientation

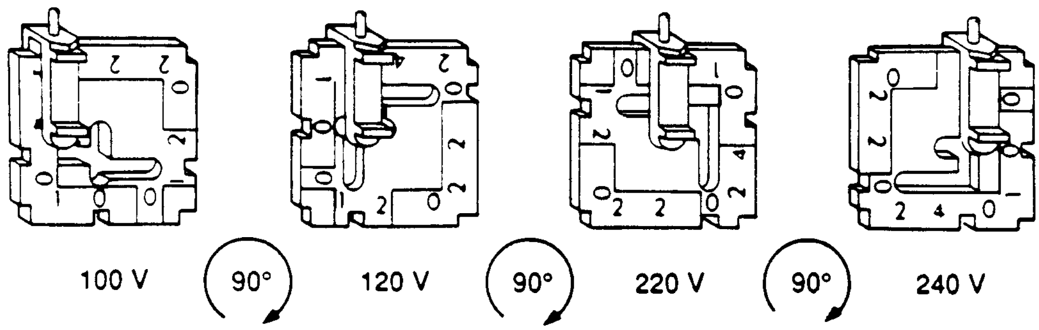

Fig. 2.4. Voltage selection and voltage card orientation.

1. Carefully remove the voltage selection card (see Figure 2.1) by pulling it straight out from the housing. Do not damage the plastic indicator pin or the metal contacts on the selector card.

2. The indicator pin can be moved across the slot located on the voltage selection card. To set the required operating voltage of the interface, orient the indicator pin on the side opposite the printed voltage on the voltage selection card (see Figure 2.4). The indicator pin must slide into the notch located on the outside edge of the voltage selection card.

3. Insert the voltage selection card into the housing, with the edge reading the desired voltage first, and the printed side facing the AC plug and off/on switch.

4. The white arrow on the indicator pin must align with the black arrow on the fuse block housing (see Figure 2.1).

5. Replace the cover, and make sure that the indicator pin shows the desired voltage.

2.3 Placement

When installing the SI, take care to allow sufficient room for ventilation. Air must be able to enter the vents at the bottom front and be expelled by the fan at the back. In general, one inch (2.5 cm) of space must be allowed on the left side.

Warning: No loose materials (such as absorbent paper) should be placed under the unit because the vent holes might block and cause overheating. Overheating may cause component failure.

2.4 Power Connections

Before connecting any cords or cables, be sure that the power switch (located at the back of the unit) is off, as indicated by the symbol 0.

Insert the power cord plug into the rear panel jack of the interface. We recommend an outlet with a grounding terminal. If this type of outlet is not available, use the grounding lug on an adaptor or the ground terminal on the rear panel of the electrical unit. The ground terminal is indicated by GND. Turn on the interface by flicking the toggle switch on the rear panel from 0 to 1.

2.5 Connecting to the Host Computer

The HRLC system interface passes information to and from the host computer via the RS-232C communications port. The interface has a 9-pin D-connector for the host computer labeled RS-232C Port 1. With the Model 500 HRLC data station, a 9-pin to 25-pin serial cable is provided. The Model 800 data station includes a 9-pin to 9-pin cable. Attach the serial cable to the interface (always 9-pin) and to the COM-1 port on the computer (either 9- or 25-pin).

The RS-232C port can be used with cables up to 100 feet (30 m) long. Extension cables are available for installations where the computer must be more than 6 feet (2 m) away from the interface (see Appendix D).

2.6 Connecting to the Gradient Module

The system interface is shipped within the gradient module, fully connected and ready to use. No additional connections are necessary.

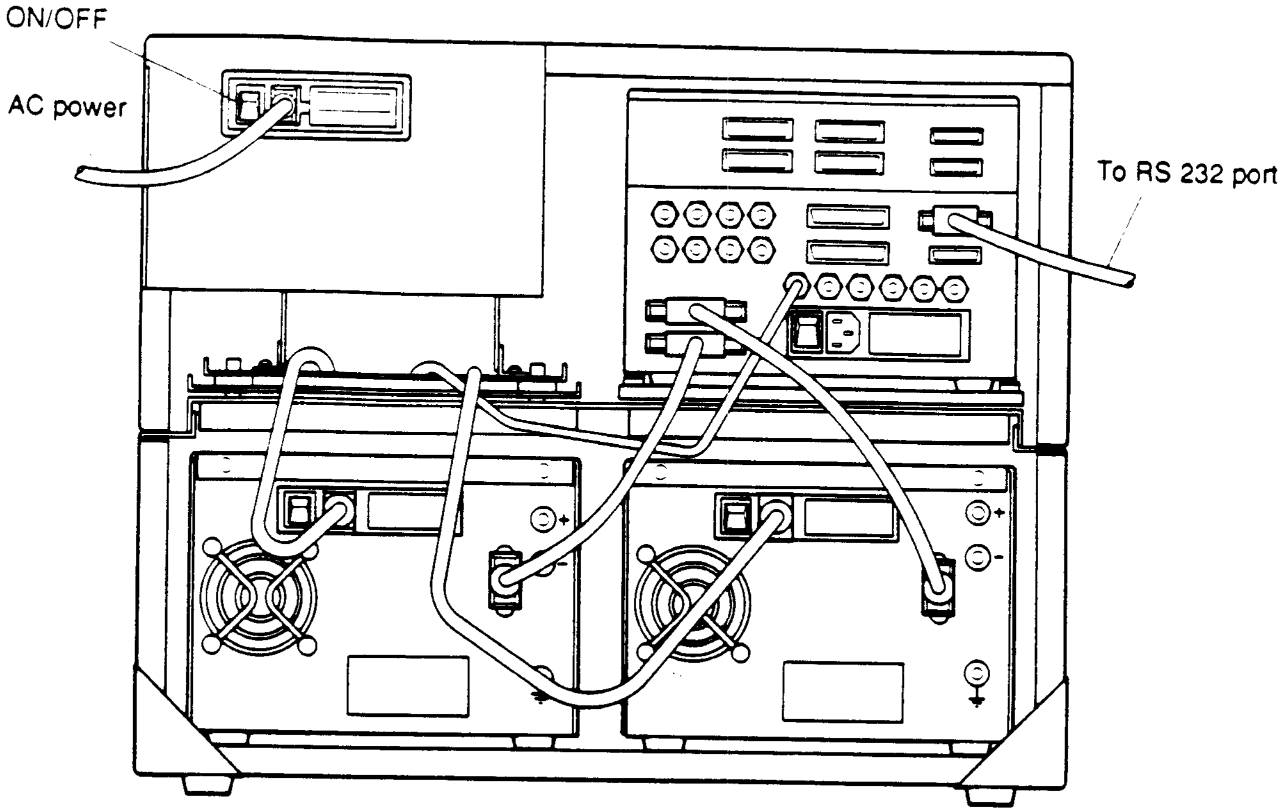

Fig. 2.5. Connections to the system interface.

2.7 Connecting to Other Components

The SI can be connected to other components, such as the MAPS 800 Preparative Module, Model AS-100 HRLC® automatic sampling system and the Model 2110 Fraction Collector. Refer to Chapter 3 for details about connection to, and operation with peripheral components.

2.8 Self-test

When the power has been turned on, the interface automatically carries out a test of its functions. During this test each of the indicator LEDs turns on, then off in sequence. The interface also executes a RAM test to insure that the read and write memory is functioning correctly.

After the test, READY should be lit, and the pump control should be in local mode. Should any other condition exist, the unit will probably not operate correctly. Contact your local Bio-Rad service center if the self-test indicates a problem. The interface's power should always be turned on and the self-test completed before initiating the HRLC software program.

Chapter 3
Operation

3.1 What the Indicator Lights Mean

All methods are run under the supervision of a host computer. The computer downloads methods and initiates the run sequence. Then the interface carries out the method. The indicator lights describe events occurring while the method is running. The lights provide status information near the system, even when the host computer is distant. A complete description of the purpose of each indicator is given below.

READY	Indicates that the system interface is prepared to receive information from the host. This indicator should be on before a run is started.
LOADING	Indicates when methods have been sent to the SI. It only turns on for only a few moments when the method is begun.
RUNNING	Indicates that a method is in progress. Since an SI can simultaneously run more than one method on more than one system, both READY and RUNNING can be lit at the same time.
DATA	Indicates that a detector card is collecting high-resolution chromatographic data.
INJECT and WAIT	Indicates that the system is waiting for an input on these lines. The input can be momentary or continuous, and can be configured to be a contact closure (TTL active low) or a contact opening (TTL active high). During these periods, the method is placed on hold. When an input (such as an injection) has taken place, the LEDs turn off.
HOLD	Indicates that the time table has paused. The system flow rate and composition are held constant during a hold, and data continues to be collected. The hold can be released by selecting CONTINUE from the software control panel. If the hold was caused by an inject or wait command, the correct input signal will restart the method.
ERROR	Indicates that a system error occurred while the method was running. System errors include activation of the stop flow mode on the pump while a method is running and system fault conditions (tripped high pressure, low pressure, or torque limit). An error immediately suspends the run in progress and stops flow on the pumps of the affected system. Eliminate the problem, and then you may resume the run.
S1, S2, S3, and S4	Depict the status of the four relays of the same names. Whenever these LEDs are lit, the corresponding relay circuits are closed. Otherwise each relay is an open circuit. These relays are normally controlled by the method, but their position can also be set manually on the software control panel.

3.2 Local and Remote Flow Rate Control

The flow rate of the Series 1350 pumps can be controlled either at the pumps or at the data station (the host computer and the HRLC software), depending on the mode of the interface system. Put the system in local mode to set the flow rate at the pumps; put it in remote mode to set the flow rate at the data station. The mode of the system can be set either at the interface or at the data station. To set the system mode at the interface, press the local/remote key in the lower left-hand corner of the interface, next to the corresponding LEDs and a membrane switch. To set the system mode at the data station, use the faceplate of the host computer running on HRLC software. For further details, refer to the HRLC Software Reference Manual. When the mode has been set, the appropriate LED, LOCAL or REMOTE, will light.

Activating either LOCAL or REMOTE will stop flow on the pumps. In order to restart the Soft-Start flow ramping of the pumps, select RUN, either from the pump control panel (on the interface) or from the faceplate of the host computer.

3.3 Use of the System Interface as an Integrator

The system interface can convert the information it collects into either high- or low-resolution digital data. Two cards on the SI perform the conversion from DC to digital data.

The detector card converts analog DC input coming from detector devices into high-resolution digital data. The card collects analog DC signals at the rate of 10 (points per sec.). The card then converts these signals into high-resolution digital values, the raw and integrated data on which chromatograms are based.

The events card converts analog DC input coming from other devices like pumps and gradient monitors into low-resolution digital data. This data can be used to monitor system temperature, conductivity, and pressure of the pumps. The events card can also be used to display, in low-resolution digital form, detector data having a limited range and fixed collection rate.

3.3.1 Hardware Connections

Hardware connections are easy to make. For each detector card, two high-resolution A to D input cables have been provided. They have a phone plug on the end of the system interface and a dual banana plug on the other end. One side of the banana plug has a tab marked GND. This side is connected to a negative polarity output. The other side is connected to a positive polarity output. To get the widest possible dynamic range for variable output detectors, select or calibrate the detector output to give 1 volt at the maximum. Attach the phone plug to the data ports on the SI.

The hardware connections for the low-resolution A to D input cables are identical to those for the high-resolution A to D input cables. A maximum of four cables is supplied with each interface and can also be used for inject input and relay output connections. Attach the phone plug to one of the monitoring inputs on the interface.

The high resolution inputs are normally used with a 0 to 1 VDC signal. Refer to Section 4.4 for adjustment procedures. Normal high-resolution range is set at 0-1 V; optional 0-10 V. Normal low-resoultion is set at 0-10 V; optional 0-1 V.

The data connections are the minimum required to use the HRLC PC integrators. Typically, an inject connection is also used. The inject connection or signal, is a contact closure activated whenever the valve is moved to the inject position. To connect the inject signal to either the inject or wait inputs use one of the phone plug cables.

Because the hardware devices can be configured by the user through the HRLC software, the hardware connection rarely needs to be changed. Several detectors can be attached to the interface, and the appropriate one can be used for specific applications. This flexibility eliminates the unpleasant tasks of frequently rewiring components in order to run different methods.

Using the interface as an integrator requires one of the HRLC data stations or a suitable IBM compatible computer, running the HRLC software. Refer to the software manual for instructions regarding the creation and running of methods.

3.4 Use of the System Interface with an Automatic Sampler

The system interface can be used with most automatic samplers. Specific cables and software drivers are available, making it easy to interface the Model AS-100 HRLC automatic sampling system and Model AS-48 Automatic Sampler.

3.4.1 Connecting to the Model AS-100 Sampler

The Model AS-100 automatic sampling system requires an optional accessory cable (see Appendix D). This cable has a 25-pin D connector on one end, and a 15-pin D-connector with two spade lugs on the other.

1. Connect the 25-pin connector to the port on the SI labeled "Auto Inject."

2. Connect the 15-pin connector to the port labeled "Sampler IO" on the rear of the AS-100. The spade lugs are labeled "START +" and "INJ". Connect these to the associated terminal lugs on the back of the AS-100 sampler.

3. Put the four dip switches on the back of the automatic sampler in the down position (LO).

4. In the system configuration screen of the AS-100 sampler, set the operating mode to remote.

The Series 800 gradient systems allow a second automatic sampler to be controlled by one SI. Follow the same procedure outlined above. At the SI, connect the D Connector to the DIGITAL I/O.

The Model AS-100 Automatic Sampler can also be connected without the optional cable. The cable requires the same four spade lug connections at the autosampler, which is connected to "inject A" on the SI. It will require a slight modification to the sample table. Refer to your HRLC Software Reference Manual.

3.4.2 Connecting to the Model AS-48 Sampler

The Model AS-48 Automatic Sampler also requires an accessory cable. This cable has a 25-pin D-connector on both ends. One end is attached to the SI port labeled AUTO-SAMPLER and the other to the Model AS-48 control unit connector CN-2. The remote switch of the control unit should be turned on, and all program dip switches (inside the control unit) should be in the standard ON position.

To use either the Model AS-100 Automatic Sampler or AS-48 Sampler with the HRLC software and the system interface, first program the injection sequence on the sampler, then press START. No injection cycle should begin. On the Model AS-100 sampler, the status should change from STOP to IDLE. On the LED screen of the Model AS-48 Sampler, START will be lit, but BUSY should remain off, and the status vial number should be 0. A method or sample table run can now be started from the HRLC software. Be certain to have the same number of iterations programmed on both the sample table data station and the automatic sampler. If fewer injections have been programmed on the sampler than on the data station, the system will hold indefinitely waiting for an injection that will not occur.

3.4.3 Connecting to Other Automatic Samplers

To use the system interface with other automatic samplers follow a similar strategy. The system works best when the SI is the master device, and the sampler is the slave. To use the system in this fashion, equip the sampler with a TTL or a contact closure input that will start each cycle, and with a similar output that will signal when an injection has occurred. The system will also work when the sampler is the master. In this arrangement, the sampler's cycle time must be greater than the method period in order to insure that the SI is waiting for the injection when it occurs. The sampler must have a TTL or a contact closure injection output signal. Refer to the instructions for the automatic sampler for further information regarding either of these procedures.

3.5 Use of the System Interface as a Pump Controller

The system interface can control up to six pumps, with a maximum of four pumps for each time-based system. The typical installation uses pump A for system 1 solvent A, pump B for system 1 solvent B, pump C for system 1 solvent C, pump D for system 1 solvent D. However, any pump device can be assigned to any system. Refer to the HRLC Software Reference Manual, Set System, for further detail. There is no need to change the pump wiring when changing system configuration. Refer to the HRLC Software Reference Manual.

Using the pump cables, the SI controls several aspects of each system: the flow rate, the choice of local or remote mode, Soft-Start flow ramping, and the stop flow mode. The SI also monitors the pump in case a system fault occurs or STOP is pressed. Use the SI local/remote key to gain control of the pump flow rate (see Section 3.2). When the pump is in remote mode, the flow rate appearing on the display is set by the data station. Otherwise, the flow rate is set at the front panel of the pump.

3.5.1 Connecting to Pumps

To add pumps to the system interface, make connections at the ports for pumps A - F on the SI. It may be helpful to label the pumps with their port connections. Each SI comes with a number of pump cables as determined by the number of pump cards within the interface.

3.6 Use of the System Interface with a Fraction Collector

The system interface can be used with most fraction collectors. Specific cables and software are available for use with Bio-Rad's Model 2110 Fraction Collector. The SI can interface with a Model 2110 Fraction Collector by means of the 2110 cable, which is connected to the digital I/O port of the SI. The system interface can control two Model 2110 Fraction Collectors if an I/O Junction Box is used as an interface between the SI and the fraction collectors. Refer to the HRLC Systems Manual to configure the system, and to Appendix D for catalog numbers.

3.6.1 Connecting to the Model 2110 Fraction Collector

The Model 2110 is connected to the SI at its Digital I/O port. A Digital I/O-to-2110 cable is required. Refer to Appendix D for catalog information.

3.7 Use of The System Interface with Optional I/O Junction Box

3.7.1 Description

The I/O junction box is used as a communication link between the system interface and auxillary components. The I/O junction box will support two Model 2110 Fraction Collectors, as well as four wait and two event signals.

3.7.2 Connecting to the I/O Junction Box

The I/O Junction box is connected to the system interface digital I/O port using the digital I/O-junction box cable catalog number 125-1648. Each junction box includes all wait and event cables. Refer to the spare parts list, Appendix D, for additional cable ordering information.

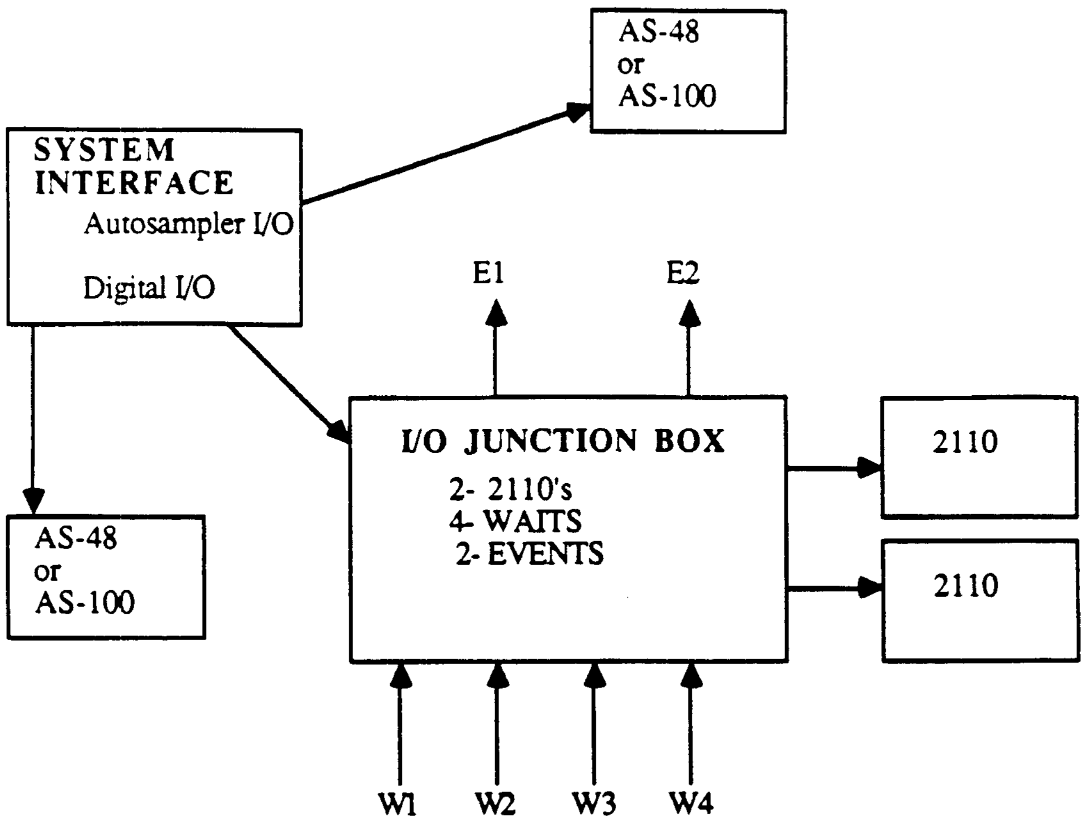

Fig. 3.1. I/O Junction box connections.

3.8 Use of SI with the MAPS 800 Preparative System

3.8.1 Description

The MAPS 800 preparative system is an integrated, preparative HPLC instrument for gram scale purification of monoclonal antibodies and other high value proteins. The system consists of a MAPS solvent delivery module which is controlled via a stand alone Model 864 system interface and a host Model 800 data station. The system interface functions as the central communications resource for the MAPS preparative system. At its fullest utility the Model 864 system interface is capable of controlling the MAPS module plus one dual pump gradient module including detectors and an automatic sampler.

3.8.2 Connecting the SI to the MAPS module

The MAPS module is connected to the system interface at the Digital I/O port and Pump A and Pump B ports. D-sub connectors connect the back panel of the system interface to the SI circuit board inside the MAPS module. The MAPS module is shipped with these cables already attached to the SI circuit board inside the MAPS module. To complete the connection to the system interface attach the 15 pin plug labelled Pump A to the Pump A port on the back of the SI and the 15 pin plug labelled Pump B to the Pump B port. Finally attach the 25 pin socket labelled Digital I/O from the MAPS module to the Digital I/O port on the back panel of the system interface.

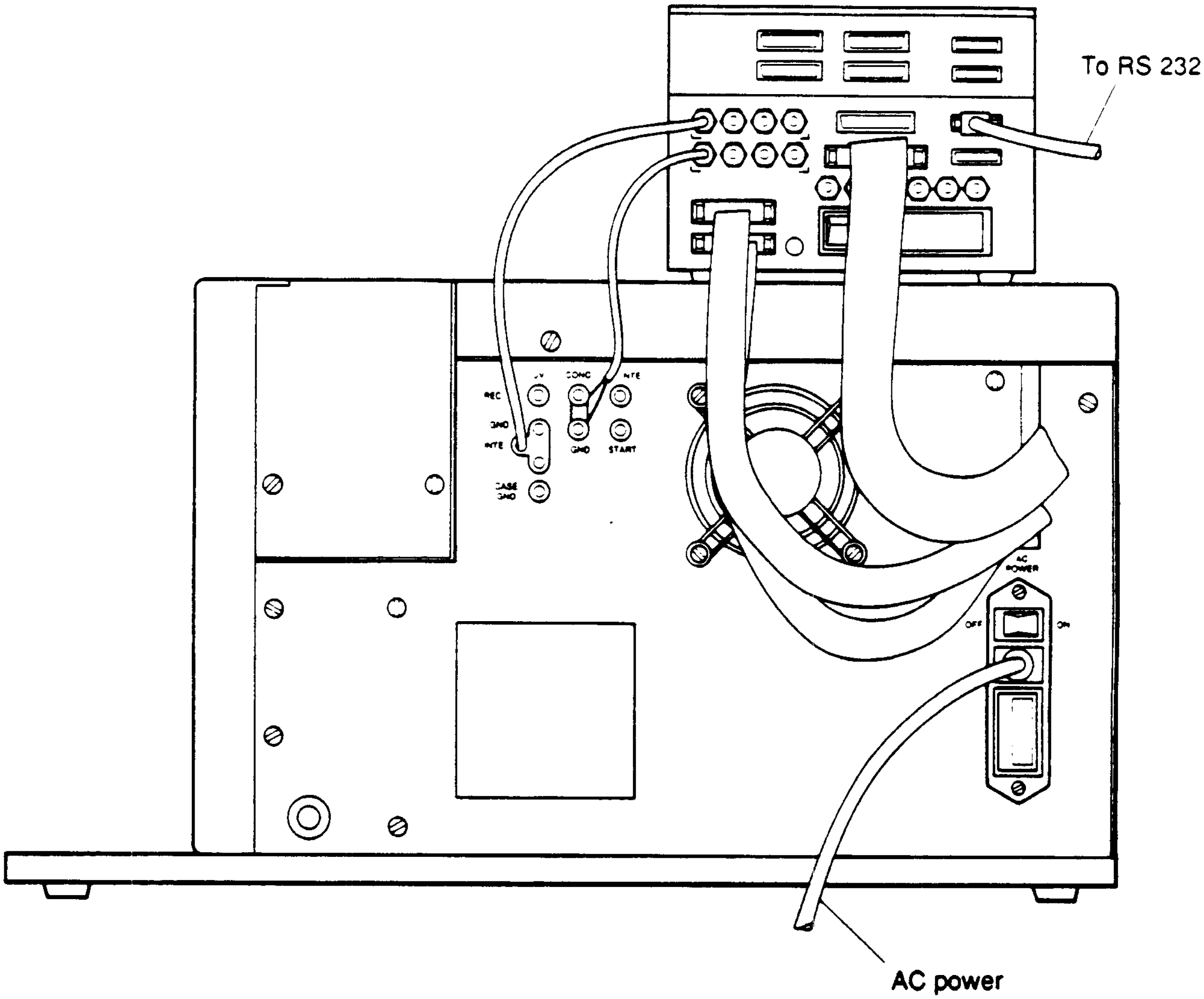

Fig. 3.2. Connections between the MAPS system and the system interface panel.

3.9 Cold Room Operation

You can operate the gradient module in a cold room as long as you turn the power on immediately upon bringing the unit into the cold room, and keep the power on the entire time it is there. If you turn the power off, immediately remove the unit from the cold room.

Note: Never allow condensation to form on the unit. It will cause malfunction of the boards.

Chapter 4
Maintenance and Service

Note: The following procedures should be performed by an instrument and service technician familiar with the HRLC software.

4.1 Routine Maintenance

No routine maintenance is necessary other than checking the analog calibration of the I/O ports. Refer to Section 4.3 for calibration test points and procedures.

4.2 What to Do If Something Goes Wrong

If you require technical assistance contact Bio-Rad Laboratories, technical service, at 1-800-4BIORAD. If you require technical assistance for instrument service, contact the instrument service department at 1-800-4BIORAD.

4.3 Calibration Test Points and Procedures

In general, calibration should be checked annually, or sooner if data acquisition or HPLC pump flow control values are clearly in error.

The following items are needed to check calibration:

1. 3 1/2 digit DVM (4 1/2 digit DVM is preferred)
2. 1 V (+ .01 V) voltage reference with 3/4" spacing dual banana jack
3. 10 V (+ .01 V) voltage reference with 3/4" spacing dual banana jack
4. Small flat-blade screwdriver
5. A cable consisting of
   - A male DB - 15P connector that plugs into the system interface pump port
   - Connectors that plug into the DVM (typically either two banana plugs or one dual 3/4" spacing banana plug)
   - Two conductor cables as needed

   To wire the cables, connect pin 12 of the male DB-15P connector to the banana plug that attaches to the positive voltage port of the DVM. Join pins one and five of the male DB-15P connector, and connect this node to the banana plug that attaches to the negative voltage input terminal of the DVM. Join together pins 14 and 15 of the male DB-15P connector. See Figure 4.1.

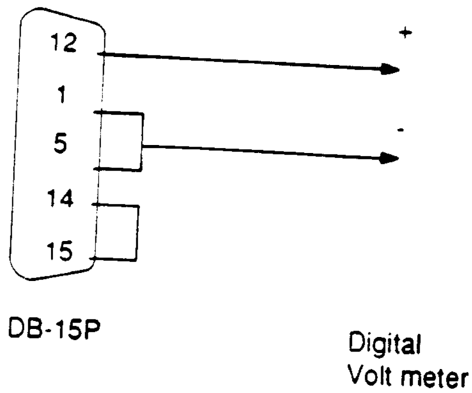

Fig. 4.1. Test Cable.

4.3.1 Calibrating the Pump Port (Analog Output Voltage)

To check the calibration of Analog Output Voltage:

1. See that the power at the system interface is off.

2. Disconnect cable from the system interface pump port to be tested and connect the cable described in Section 4.3 number 5 to the SI port and to the DVM.

3. Turn on the system interface and let it warm up for 10 minutes. Turn on the data station and run the HRLC program. Run the solvent delivery section from the instruments menu.

4. Select 0% and 10 ml/min for the port being tested. The DVM should read 0.000 V +/- 0.010 V. There is no output zero adjustment.

5. Select 100% and 10 ml/min for the port being tested. The DVM should read 9.997 +/- 0.010 V. Output span adjustment is possible in the field.

If both steps four and five yield values beyond specified limits, make sure that the cable connecting the pump port of the system interface to the DVM conforms to the description in Section 4.3, Item 5. If the cable does conform to the description, and performing step four still leads to values outside of specifications, return the unit for service. If only step five yields values outside of specifications, the following adjustments may be made:

To adjust the calibration of analog output voltage:

6. Gain access to the pump port circuit cards of the system interface by unscrewing the retainer screws in the upper left and right corners of the SI front panel and letting the panel drop forward on its hinge.

7. Perform steps 1 through 5 of the calibration check.

8. Pump port cards are located in slots 4, 5, 6, or 7. The slot numbers are visible on the card cage above the card ejectors. Locate the appropriate pump circuit card and adjustment pot from the following table:

To Adjust	Locate Cable Marked	Adjustment Pot Is
PUMP A	PMP6-J1	R5
PUMP B	PMP6-J2	R7
PUMP C	PMP5-J1	R5
PUMP D	PMP5-J2	R7
PUMP E	PMP4-J1	R5
PUMP F	PMP4-J2	R7

The cables are marked where they plug into the circuit cards. Adjustment pot R5 is the topmost pot, and adjustment pot R7 is underneath it. There is another pot beneath R7 (called R8). Be careful not to change its adjustment. See Figure 5.6 for details.

9. Adjust the appropriate pot (R5 or R7) for the calibration required in step 5 of the calibration check. Use the small flat-bladed screwdriver.

10. If adjustment cannot bring the output within specifications, set the pot adjusted in step 4 above the middle of its range and adjust R8 to bring the output within specifications. If this is done, the pot (R5 or R7) for the other pump port controlled by the board must be adjusted as described in 4 above. If the pump ports cannot be adjusted, the unit should be returned for service.

4.3.2 Verifying the Calibration of the Data Port

1. Turn on the SI and data station and run the HRLC Program. Run a method that will generate a strip chart display that plots the data channel under test.

2. Plug a cable into the data port under test and apply 0 V to the port. The strip chart should plot between 8% and 14%.

3. Apply 1 V (+/- . 01 V) for inputs configured for a 1 V span, or 10 V (+/- .01 V) for inputs configured for a 10 V span. The strip chart should plot between 88% and 100%.

If steps 2 and 3 result in values beyond the limits specified, and the connections, cabling and voltage reference are in good order, the unit is defective and should be returned for service. No adjustment is possible outside the factory.

4.3.3 Calibrating the Monitor Port (Analog Input Voltage)

1. Turn on the system interface and data station and run the HRLC program. Run a method that will generate a strip chart display plotting the monitor channel under test.

2. Plug a cable into the data port under test and apply 0 V to the port. The strip chart should plot between 0 and 2%. If the strip chart reads outside this range, the unit is defective and should be returned for service.

3. Apply 1 V (+/- .01 V) (for channels configured for a 1 V span) or 10 V (+/- .01 V) (for channels configured for 10 V span). The strip chart should plot between 98% and 100%.

4. If the plot in 3 is outside this range, open the front panel. With the port configured for a 10 V span (refer to the adjusting input range of the system interface data, monitor inputs), adjust the topmost adjustment pot (R8) of the circuit card in slot 2, while performing step 3 for a10 V span.

5. If the 1 V span is being used, configure the port for a 1 V span, and adjust the bottommost adjustment pot of the leftmost circuit card (R14), while performing step 3 for a 1 V span.

If the necessary adjustments cannot be performed successfully, the unit is defective and should be returned for service.

4.4 Configuring the Data and Monitor Port Input Ranges

Data and monitor analog input ports of the system interface are field-configurable for the ranges between 0-1 V or 0-10 V.

4.4.1 Configuration of DATA Analog Input Port Range

1. See that the power of the system interface is turned off.

2. Unscrew the retaining screws on the top left and right of the system interface front panel, and fold the front panel down to gain access to the circuit cards.

3. Locate the circuit card for the data input to be configured. The data input card for data 1 and 2 is located in slot 8 of the card cage, and the card for data 3 and 4 is located in slot 7. The slot numbers are marked above the cards.

4. Unplug the cables for the card to be configured and remove the card from the system interface. It may be necessary to remove cables from other cards in the process; keep track of the cables. If cables are crossed, refer to Figure 5.5.

Note: To prevent possible damage to electronic cards, take care to avoid static electricity by wearing ground straps on the wrist and by placing the card on a grounded workbench.

5. Orient the card so that the component side faces you, and the lettering on the card surface reads right side up. Locate configuration jumper group W2 in the upper right area near the card ejector (Figure 4.2 System Interface Data Card). The right-hand half of W2 is for data 1 or 3; the left-hand half is for data 2 or 4.

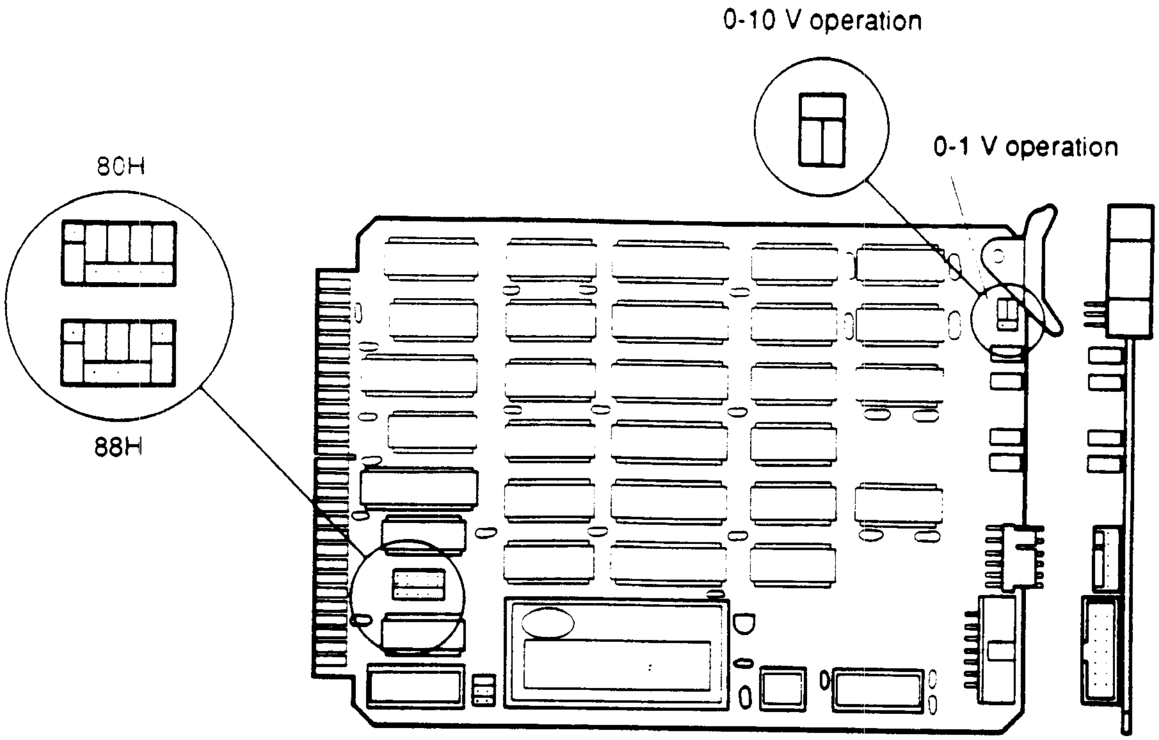

Fig. 4.2. System Interface A/D card.

6. To configure for the 0-1 V range, place a jumper plug in the upper half of W2; to configure for the 0-10 V range, place a jumper plug in the lower half of W2. Units are shipped from the factory configured for the 0-1 V range.

7. Install the card in the slot of the card cage it was removed from. Reconnect at their original locations all cables removed in step 4.

8. Fold the front panel up and tighten the retaining screws snugly. Avoid pinching cables between the panel and chassis.

4.4.2 Configuring the Monitor Analog Input Port

1. See that the power of the system interface is turned off.

2. Unscrew the retaining screws on the top left and right of the front panel of the system interface, and fold the front panel down to gain access to the circuit cards.

3. Locate the monitor input circuit card in slot 2. Remove the two flat cables on the edge of the card and slide the card out of the card cage halfway. The other flat cables plugged into the card may be left in place. Follow static electricity precautions.

4. View the component side of the card with the lettering on the card surface reading right side up. Locate configuration block W3 in the lower right corner near connector J5.

5. Each post in the configuration block is numbered between 1 and 12 as marked on the card. To configure each port, place jumper plugs to connect post pairs per the following:

Jumper Plug Location:

DATA Port	0-1 V	0-10 V
1	1-2	2-3
2	4-5	5-6
3	7-8	8-9
4	10-11	11-12

Units are shipped from the factory with data ports 1 and 2 configured for 0-1 volt, and data ports 3 and 4 configured for 0-10 volts.

6. Install the card in the slot of the card cage that it was removed from, and reconnect at their original locations all cables removed in step 3.

7. Fold the front panel up and tighten the retaining screws snugly. Avoid pinching cables between the panel and chassis.

Chapter 5
Adding Hardware to the System Interface

The System interface was designed to grow as the chromatographers needs grow. In the maximum configuration, the SI can support six pumps, four high resolution data channels, and four low resolution channels. Features can be added simply by plugging printed circuit boards into the computer bus.

All systems come standard with the ability to control two pumps. Each additional pump controller card allows for control of two more pumps.With these 3 pump controller cards the SI can control six pumps.

Each A/D convertor card provides two channels of high resolution data for a maximum of four high resolution channels.

All 800 series HRLC systems have a pump expansion panel on the rear of the SI. To convert a 500 series SI into an 800 series SI requires the addition of this panel, which has the connections for pumps C through F.

Caution: Electronic assemblies have static sensitive electronic components. Avoid static electricity by working only at a grounded workstation.

DISCONNECT ELECTRICAL POWER BEFORE WORKING ON ANY EQUIPMENT.

5.1 Adding a Pump Expansion Panel

A pump expansion panel is added to upgrade from a 500 series system to an 800 series system. This panel allows the HRLC system to control more than two pumps. If your SI already has plugs on the back for pumps C through F, the expansion panel has already been installed.

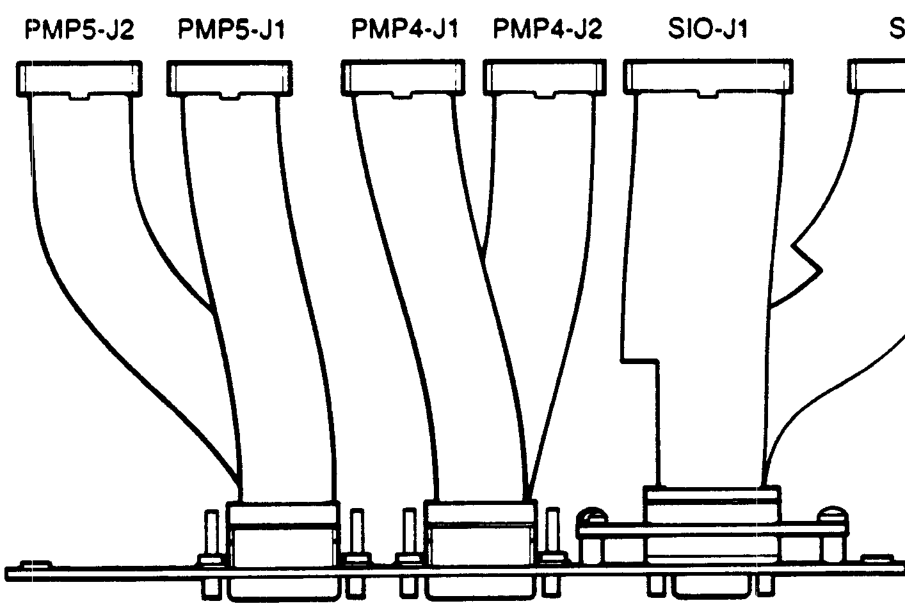

Fig. 5.1. Expansion panel showing cables.

Remove the top cover of the system interface by removing the four phillips head screws on each side of the instrument and the two phillips head screws on the top rear of the unit. Remove the blank panel on the top rear of the SI. Install the new panel and route the cables as shown in Figure 5.2.

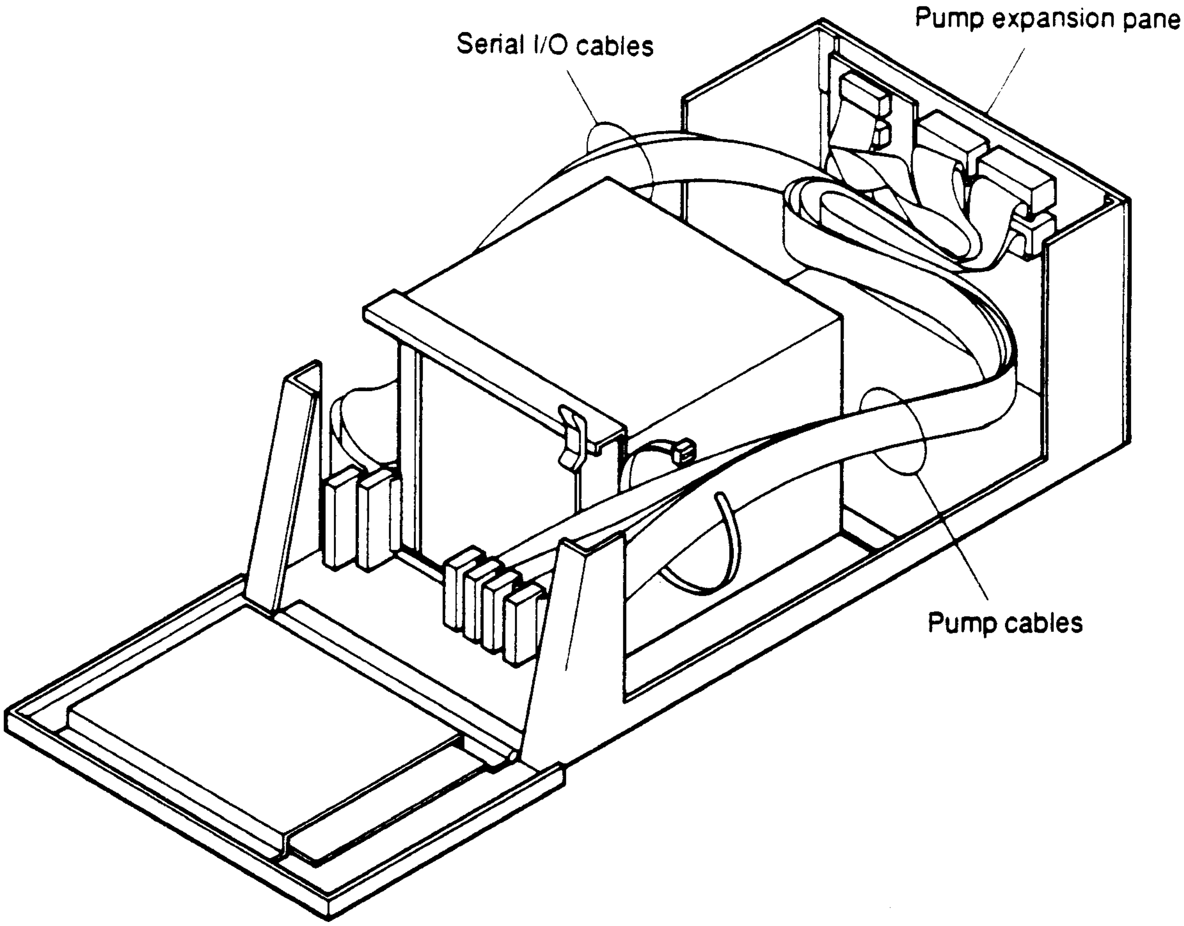

Fig. 5.2. SI with top removed.

5.2 Adding an Analog to Digital (A/D) Card

The analog to digital convertor cards are designed for slots 7 and 8 of the SI. The first A/D card (slot 8) should be configured for address 80, and the second card (slot 7) should be set to address 88. Addresses are selected by placing shorting jumpers on the printed circuit board.

1. Before installing the A/D cards open the front of the SI by loosening the two holding screws on the front panel. The top will fold forward and expose the card cage. Determine if you already have an A/D card installed.

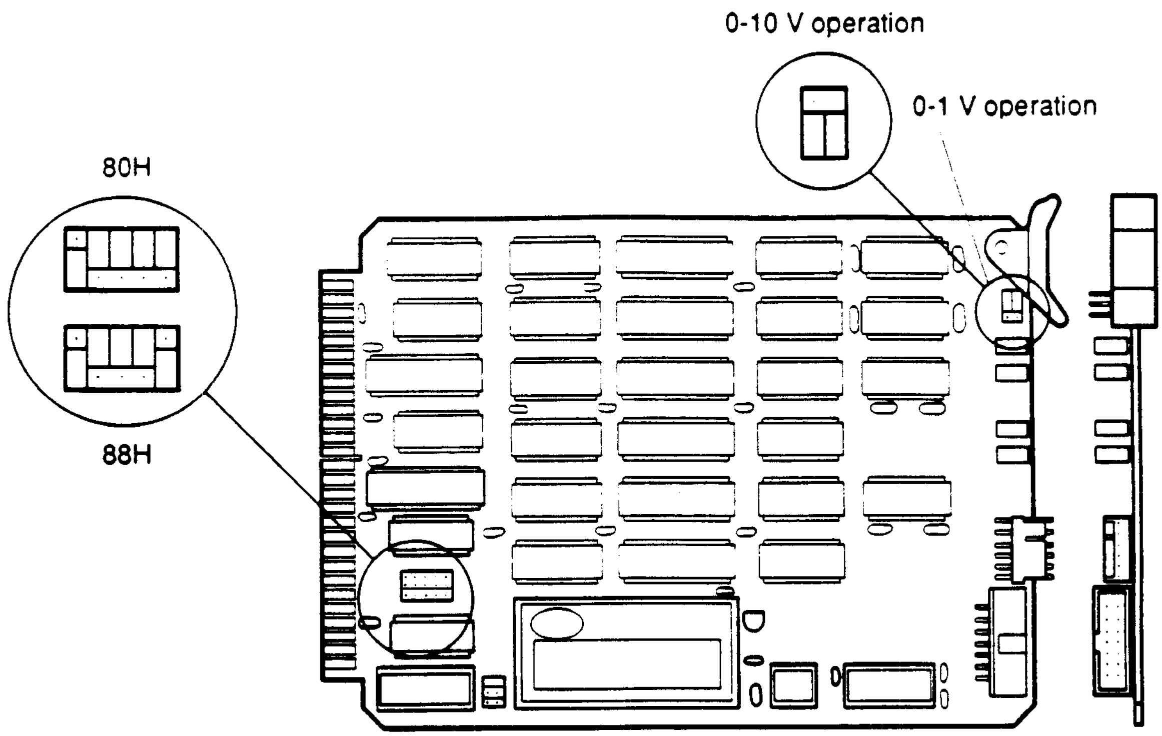

Fig. 5.3. A/D card top and end views.

2. If you have no A/D card, configure the jumper block for address 80 according to Figure 5.3. If you already have one A/D card configure the jumper block for address 88.

3. The analog to digital convertor is capable of accepting two different voltage range inputs, either 0-10 volt, or 0-1 volt. Normally the input should be configured to 0-1 volt. This is done by placing jumpers on the A/D circuit board. Refer to Figure 5.3 and section 4.4.1 (Configuration of DATA Analog Input Port Range) for details.

4. The first card was designed to be inserted into slot 8 in the SI, and the second card was designed for slot 7. To do this any card already installed into that slot will have to be moved. To maintain future system compatibility all the pump cards should be shifted one slot over to the left to make room for your additional A/D card.

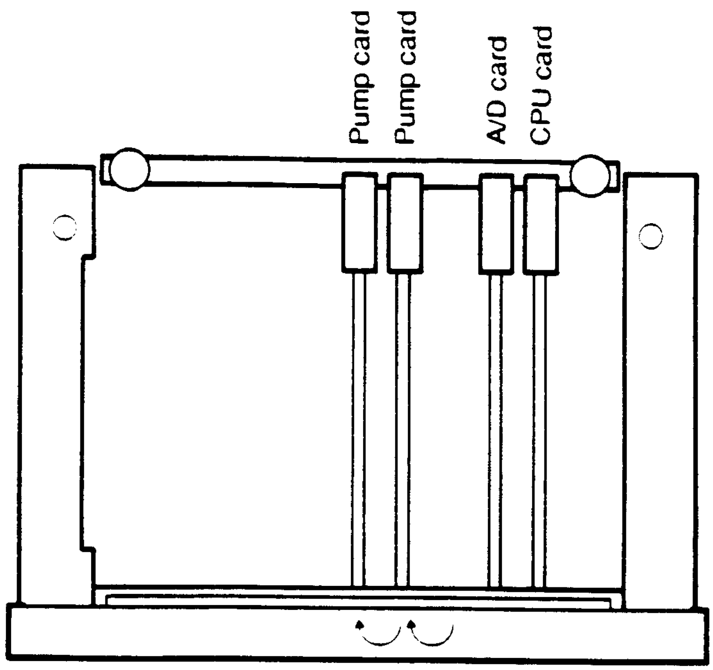

Fig. 5.4. Shifting pump cards to the left.

5. Before moving any cards, care should be taken to ensure the correct cables will be connected back where they belong. Figure 5.5 shows the card cage layout and cable connections. Refer to this drawing if cables get crossed.

CARD SLOT

MODEL	1	2	3	4	5	6	7	8	9
864		MF IO CARD		PUMP 70H PMP4-J1 PMP4-J2	PUMP 68H PMP5-J1 PMP5-J2	PUMP 60H PMP6-J1 PMP6-J2	A/D 88H AD7-P2 AD7-J2	A/D 80H AD8-P2 AD8-J2	CPU CPU-J3 CPU-J1 CPU-J2
862		MF IO CARD			PUMP 70H PMP4-J1 PMP4-J2	PUMP 68H PMP5-J1 PMP5-J2	PUMP 60H PMP6-J1 PMP6-J2	A/D 80H AD8-P2 AD8-J2	CPU CPU-J3 CPU-J1 CPU-J2
844		MF IO CARD			PUMP 68H PMP5-J1 PMP5-J2	PUMP 60H PMP6-J1 PMP6-J2	A/D 88H AD7-P2 AD7-J2	A/D 80H AD8-P2 AD8-J2	CPU CPU-J3 CPU-J1 CPU-J2
842		MF IO CARD				PUMP 68H PMP5-J1 PMP5-J2	PUMP 60H PMP6-J1 PMP6-J2	A/D 80H AD8-P2 AD8-J2	CPU CPU-J3 CPU-J1 CPU-J2
822		MF IO CARD					PUMP 60H PMP6-J1 PMP6-J2	A/D 80H AD8-P2 AD8-J2	CPU CPU-J3 CPU-J1 CPU-J2

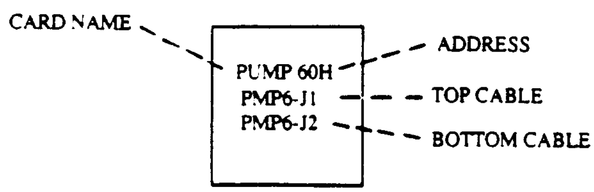

Fig. 5.5. System interface card cage layout.

6. Carefully align the card in the card slot and slide it into the rack. When the card edge contacts the connectors in the rear of the system you will have to push firmly to get it to seat properly. DO NOT FORCE THE CARD INTO POSITION. It should be a snug fit.

7. Connect either cable AD8-P2 or AD7-P2 (see Figure 5.5) to the top of the A/D card, and cable number AD8-J2 or AD7-J2 to the bottom connector on the A/D card. If you are upgrading from a 520 system, remove the jumper plug on the CPU card and discard. Attach cable # CPU-J3 to the CPU card and then to your new A/D card.

8. To verify correct operation of the two new analog channels, reconfigure the HRLC system to include these two channels, and run a method with them assigned to a strip chart. See section 6.2 in the HRLC Software Reference Manual for details. If you have a detector input connected to the two new data channels, they will be plotted on the strip chart when your method is run.

5.3 Adding Pump Cards

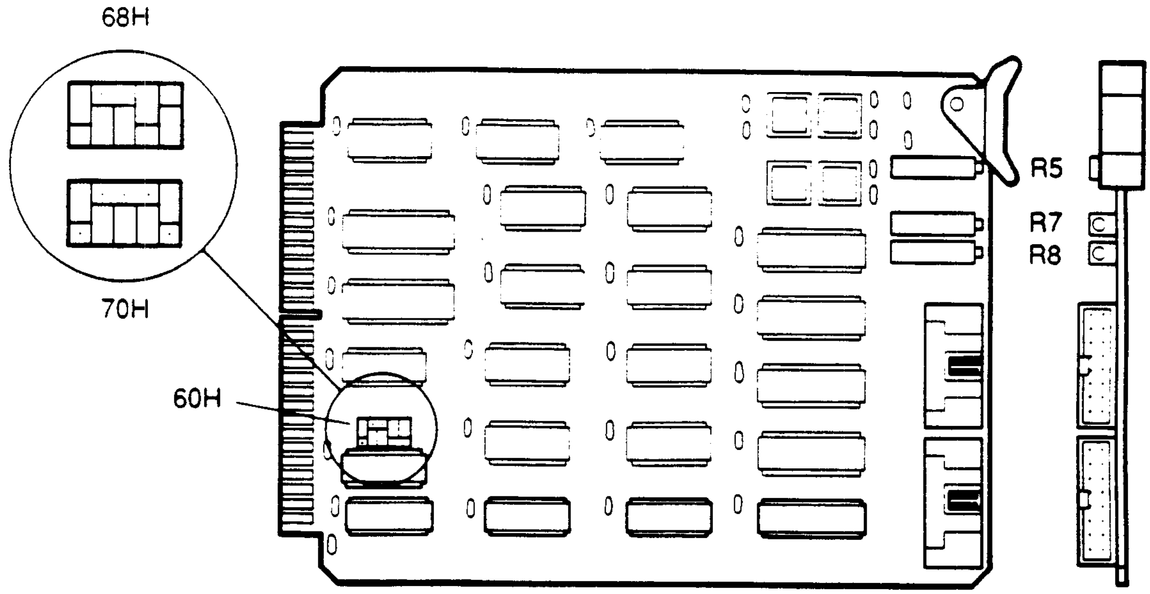

Fig. 5.6. Pump card top and end view.

1. Before adding a pump card the jumpers on the printed circuit card must be properly set to operate with your system. All system interfaces come with at least one pump controller card. This card is assigned to address 60 (hex). The second pump controller card must be assigned to address 68 (hex), and the third must be assigned to address 70 (hex). Address selection is accomplished by placing jumper plugs on the circuit board. To set the card address jumpers for the pump controller cards refer to Figure 5.6.

2. Once the address jumpers have been set, the pump controller card can be inserted into the card cage. Open the front of the SI by loosening the two holding screws on the front panel. The front panel will fold forward and expose the card cage.

3. The card is designed to be inserted into the system interface in the open slot farthest to the right. Depending on your current system configuration this will be either slot 4, 5, or 6. See Figure 5.5 for card configuration. Carefully align the card in the card slot and slide it into the rack. When the card edge contacts the connectors in the rear of the system you will have to push firmly to get it to seat properly. DO NOT FORCE THE CARD INTO POSITION. It should be a snug fit.

4. The card was fully tested in the factory before shipment and no additional alignment should be required. In the event alignment is required refer to section 4.3.1, Calibrating the Pump Port.

Appendix A
List of Figures and Tables

Figures

Tables

Appendix B
Configuration and Performance Specifications of SI Hardware

Listed below are the performance specifications for each standard system interface. Each SI can handle a maximum of six pumps and four detectors.

1. Hardware Configuration

General Specifications:

4 - or 6- MHZ Z-80-based STD - Bus microcomputer, with ability to incorporate:

- 4 RS-232C serial ports
- 6 1350 HPLC pump control ports
- 2 AS-48/AS-100 autosampler ports or
- 1 AS-48/AS-100 autosampler port and 2 FC-2110 Fraction Controller ports with 4 contact closure/TTL inputs
- 1 manual inject and 1 wait contact closure/TTL input
- 4 high-resolution analog data-acquisition inputs
- 4 low-resolution analog data-acquisition inputs

Other equipment:

- Multifunction 6.0 MHz Z80 processor
- Multifunction I/O Card
- 2 Channel Pump Controller Card (up to 3)
- 2 Channel A/D Card (up to 2)
- Status Display Front Panel Card
- Power Supply
- Cables:
  Serial cable
  (2-6) Pump control cables
  (2-4) Detector Cables
  Inject/Wait cables
  General I/O cables
- Standard Rear Panel
  8 analog input connectors
  4 relay outputs
  2 TTL inputs
  Autosampler connector
  Digital connector
- 2 RS-232 connectors
- 2 1350 Pump control connectors
- Optional rear panel extension
  4 Additional 1350 pump control connectors
  2 Additional RS-232 connectors
- Standard Card Rack
- Cabinet

2. Performance Specifications

2.1 Power Requirements

Line Voltage:	95-135 VAC or 190-270 VAC, user-selectable, at 47-63 Hz
Line Current:	At 115 VAC, less than 0.9 A; at 230 VAC, less than 0.5 A
Fuse ratings:	2 A at 95-135 VAC, 1 A at 190-270 VAC

2.2 Ambient Temperature Range

0 degrees centigrade -40 degrees centigrade (non-condensing).

2.3 Digital Inputs

May be driven by TTL or 5V CMOS with 0.5 mA current sink (typical of all TTL, HCMOS logic and 4000 CMOS logic with TTL drive capability) and by contact closure outputs.

2.4 Digital Outputs

5V HCMOS with 2 mA minimum current sink/source capability; will drive all TTL and 5V CMOS logic inputs.

2.5 Low-Resolution Analog Data Inputs

12-bit resolution, 0.05% linearity, conversion rate of 1 sample per second. Each channel can be hardware-configured for 0-1 V or 0-10 V input span. Inputs are single-ended and are referenced to an STD-Bus analog ground.

2.6 High-Resolution Analog Data Inputs

Resolution depends on sample rate. At 10 samples/sec, resolution is 1 part in 200,000 (between 17 and 18 bits). At 5 samples/sec, resolution is 1 part in 500,000 (19 bits). At 2 samples/sec, resolution is 1 part in 1,000,000 (20 bits). Linearity is 0.05% worst case, 0.02% typical. Each channel can be hardware-configured for 0-1 or 0-10 V input span with 10% underrange and overrange. Inputs are differential with a (±) 10 V common mode input range. The analog front end is isolated up to 300 V.

2.7 1350 HPLC Pump Control Port Analog Voltage Output

12-bit resolution, adjustable to accuracy corresponding to ± 1 microliter in 1350 applications. Output span is 0 to 10 V.

Appendix C
Pin Assignments for Rear Panel Connections

SI Rear Panel: Assignment of Signals to Rear Panel Connectors

1. RS-232C Ports 1, 2, 3, 4
   Connector is female DB-9 (DB-9S)

Pin	Signal
1	no connection
2	RxD
3	TxD
4	DTR
5	Signal Ground
6	DSR
7	RTS
8	CTS
9	no connection

2. Pumps A, B, C, D, E, F
   Connector is female DB-15 (DB-15S)

Pin	Signal
1	Digital signal ground
2	System error out
3	Stop flow out
4	System fault in
5	Analog signal ground
6	Running in
7	no connection
8	no connection
9	no connection
10	Run out
11	Stop flow in
12	Analog flow control voltage out
13	Local/remote out
14	Loopback from 15
15	Loopback from 14

3. Autosampler I/O
Connector is male DB-25 (DB-25P)

Pin	Signal
1	Sample vial number D0 (LSB) in
2	Sample vial number D1 in
3	Sample vial number D2 in
4	Sample vial number D3 in
5	Sample vial number D4 in
6	Sample vial number D5 in
7	Sample vial number D6 in
8	Inject in
9	Load sample out
10	no connection
11	no connection
12	no connection
13	no connection
14	ground
15	ground
16	ground
17	ground
18	ground
19	ground
20	ground
21	ground
22	ground
23	ground
24	no connection
25	no connection

4. Digital I/O
Connector is male DB-25 (DB-25P)

Pin	Signal (Automatic Sampler)	Signal (Fraction Collector)	Signal (MAPS)
1	Sample vial no. D0 in	Wait 1 in	Inject in
2	Sample vial no. D1 in	Wait 2 in	no connection
3	Sample vial no. D2 in	Wait 3 in	no connection
4	Sample vial no. D3 in	Wait 4 in	no connection
5	Sample vial no. D4 in	FC1 Ready in	FC waste/collect out
6	Sample vial no. D5 in	FC2 Ready in	Sample pump sw out
7	Sample vial no. D6 in	no connection	Prep inject out
8	Inject	no connection	FC advance out
9	Load sample out	FC1 ext adv out	Initialization out
10	no connection	FC2 ext adv out	AC power control out
11	no connection	no connection	Solvent A valve out
12	no connection	no connection	Solvent B valve out
13	no connection	no connection	no connection
14	ground	ground	ground

Pin	Signal (Automatic Sampler)	Signal (Fraction Collector)	Signal (MAPS)
15	ground	ground	ground
16	no connection	no connection	no connection
17	no connection	no connection	no connection
18	no connection	no connection	no connection
19	no connection	no connection	no connection
20	no connection	no connection	no connection
21	no connection	no connection	no connection
22	no connection	no connection	no connection
23	no connection	no connection	no connection
24	no connection	no connection	no connection
25	no connection	no connection	no connection

5. Data 1, 2, 3, 4
Connector is 3-circuit 1/4" phone jack

Pin	Signal
tip	Analog voltage input +
ring	Analog voltage input -
sleeve	shield

6. Monitor 1, 2, 3, 4
Connector is 3-circuit 1/4" phone jack

Pin	Signal
tip	Analog voltage input
ring	Analog voltage return
sleeve	shield

7. Inject, Wait
Connector is 3-circuit 1/4" phone jack

Pin	Signal
tip	Input
ring	Return
sleeve	shield

8. S1, S2, S3, S4
Connector is 3-circuit 1/4" phone jack

Pin	Signal
tip	Relay contact
ring	Relay contct
sleeve	shield

Appendix D
Accessories and Spare Parts

Catalog Number	Description
125-1170	Bus Mouse
125-1172	Mouse Pad
125-1574	Microsoft Windows
125-1575	Microsoft Windows-386
125-1576	Bus Mouse and Windows
125-1577	Bus Mouse and Windows-386
125-1578	MS Windows - 286
125-1579	Bus Mouse and Windows - 286
125-1580	Ternary Adapter Kit-SS
125-1582	Ternary Adapter Kit-TI
125-1585	Quaternary Adapter Kit-SS
125-1587	Quaternary Adapter Kit-TI
125-1589	500 PC Integrator Software
125-1590	Series 500 HRLC Software
125-1591	800 PC Integrator Software
125-1592	Series 800 HRLC Software
125-1594	Series 500 Solvent Only HRLC Software
125-1600	502 Interface
125-1610	804 Interface
125-1614	864 Interface
125-1620	Dual A to D Kit
125-1622	A to D Cable, High Resolution
125-1624	A to D Cable, Low Resolution
125-1625	SI Data Expansion Kit
125-1630	Dual Pump Kit
125-1632	Pump Cable
125-1635	SI Pump Expansion Kit
125-1640	Communications Kit
125-1642	RS-232C Serial Cable, 9-25
125-1644	RS-232C Serial Cable, 9-9
125-1652	AS-48/Digital I/O -Autosampler I/OCable
125-1654	AS-100/Digital I/O -Autosampler I/OCable
125-1646	I/O Junction Box

Catalog Number	Description
125-1618	Manual Injector Cable
125-1648	Digital I/O to Junction Box Cable
125-1658	Digital I/O to 2110 Cable
125-1650	Relay and Wait Cable to Junction Box
125-1646	Junction Box to 2110 Cable
125-1656	Model 2110 Fraction Collector, Remote Cable
155-2009	MAPS-System Interface Cable
155-2012	SI-MAPS Pump Cable
125-1700	Model 8001 Gradient Module, 110/120/220/240 V
125-1701	Model 8001T Gradient Module, 110/120/220/240 V
125-1702	Model 8004T Gradient Module, 110/120/220/240 V
155-2013	MAPS 800 10 ml/min Module, 110 V
155-2014	MAPS 800 10 ml/min Module, 220 V
155-2015	MAPS 800 40 ml/min Module, 110 V
155-2016	MAPS 800 40 ml/min Module, 220 V

Appendix E
Error Codes and Messages

Problem:	HRLC system informs user that the pump is not connected.
Solution:	A pump port configured in "HRLC" (software) may not be physically present. Cable between the system interface and the pump may be unconnected or defective. Cable inside SI for pump port may be disconnected from circuit card or defective. Expansion kits may not have been properly installed.
Problem:	HRLC system informs user that the pressure limit has been exceeded.
Solution:	Pump system fault has been tripped. Reset it at the pump control panel.
Problem:	Pump control panel status lights are lit, and pump system fault alarm is sounding.
Solution:	The AC power of the system interface is off, and the interface is connected to a pump with AC power on. Turning on SI power will result in normal pump control status.
Problem:	Strip chart displays baseline only for data and monitor inputs.
Solution:	The run may be generating peaks smaller than the strip chart can display at its current vertical scaling. Select a smaller vertical scale (less than 100%). Or, the input selected for display may not have an analog voltage source connected to it. Or, unconnected monitor inputs may be producing erratic data. Check card configuration against instructions when expansion kits are being installed.
Problem:	Automatic sampler sample vial number acquisition and automatic sampler control are not working correctly.
Solution:	Check settings of the automatic sampler control signal configuration switches against settings recommended for use with automatic samplers connected to a system interface. Check cable connections.
Problem:	SI front panel LEDs do not light when power is turned on.
Solution:	Check line cord connections and check SI power fuse. SI may have been damaged by attempting to run a 110 V configured unit at 220 V.
Problem:	HRLC informs user of failure to communicate with the system interface.
Solution:	Cable between the interface and data station may be improperly connected or defective. Or, cables inside SI may not have been connected correctly during field configuration or expansion. Or, SI power may not be on; check front panel LEDs.
Problem:	Model 2110 Fraction Collector advances without a running method.
Solution:	Model 2110 Fraction Collector will advance once SI power has been turned on.